NET-V4: Technical Evidence Companion

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This companion provides complete protocols, source checks, seed-level results and adverse utility findings for NET-V4. This supplement retains the detailed evidence accompanying the main manuscript. Its section and table numbers are local to this document; the final main manuscript and its appendices govern the current scope and conclusions. The resource's English reader guide states model coverage, commands and output interpretation. Its construction documentation binds profile, AR(1), pulse and recovery settings to the pinned legacy source; it does not recover the missing historical raw-source pool. The original 256-context adapter remains center-only. The separate NET-V4-MV adapter supplies all nine numeric input slots for joint prediction, with profile-similarity sequences rather than measured road adjacency.

A Original snapshot characterization

A.1 Event-window scores mostly evaluate post-pulse targets

The full masks reveal no pre-pulse target points in either test partition. Of the 40,476 points in event windows, only 3,405 (8.41%) are pulse marked; the other 37,071 (91.59%) follow the pulse. Even among windows with future marks, only 12.49% of their 27,252 target points are pulse marked. The same pattern appears descriptively in all 20,000 event rows: 20,299 of 240,000 future points (8.46%) are marked. The test population and the full resource must not be conflated.

Coverage is also concentrated near the prediction cutoff (Figure 1). Pulse counts at steps 1–5 are 2,271, 1,129, 3, 1 and 1; steps 6–12 have none. Therefore 3,400 of 3,405 marked targets (99.85%) occur at nominal horizons of 15 or 30 minutes. A low marked-step MAE supplies little evidence about sustained pulse prediction at longer horizons. Reweighting the available scores cannot supply absent long-horizon pulse observations.

STID has the lowest post-pulse mean MAE among these configurations: 10.118 ± 0.214 in the generated variant and 10.307 ± 0.083 in the mixed variant (training-seed standard deviations). Its generated-snapshot event MAE is exactly the mixture $0.084124 \times 4.822711 + 0.915876 \times 10.118190 = 9.672714$ before displayed rounding. Thus the whole-event result places most scoring weight after the pulse; it is not interchangeable with its marked-step result. All configurations have higher post-pulse than pulse MAE, but this does not identify a recovery effect: the phase comparison also changes forecast distances and samples. The paired construction diagnostic below addresses a different, controlled question.

A.2 Phase-based model selection has limited utility

Validation-based pulse selection chooses center ridge on the generated snapshot: test pulse MAE falls from 4.822711 to 4.790698, but overall MAE rises from 11.498972 to 14.395967. Other criteria, all mixed-snapshot criteria and the single-seed sensitivity do not support a stable selection benefit. The exposed-data protocol and full results are retained (Appendix K).

A.3 The absolute-error association survives

Every configuration in Table 1 has lower event than control MAE in both variants. Last value's differences are −7.901 generated and −39.827 mixed. Source asymmetry, generation, scale and reconstructed metadata preclude attributing this association to real emergencies making traffic easier to forecast.

The legacy generator also supplies a concrete reason to avoid assigning the observation solely to real traffic regimes. Before clipping, its post-event recovery is

$$p_i = (1 - r_i)a + r_i v_i, \qquad r_i = 1 - e^{-(i-e)/\tau}, \tag{1}$$

where a is the final perturbed event value, e is the first recovery index, and v_i is the base generated volume. With a fixed, a perturbation of v_i is multiplied by r_i. At $i = e$, that coefficient is zero. Six finite-difference checks of the extracted legacy function agree with this algebra to 10^{-10}. This is a generator property and a constructed code check, not a decomposition of the measured error gap. It establishes a construction mechanism that must be considered before attributing predictability to real event timing.

A.4 Metric and model-ordering boundaries

Linear drift and both ridge variants have positive event-minus-control WMAPE differences on the generated snapshot despite negative MAE differences; all six mixed-snapshot configurations retain lower event WMAPE. Appendix G retains the complete table. Different denominators answer different questions.

Among the original six configurations, STID has the lowest overall MAE: 11.499 ± 0.029 generated and 26.172 ± 0.358 mixed. Its generated pulse MAE (4.823) is slightly above local-9/center ridge (4.774/4.791), but its seed standard deviation 0.110 exceeds both gaps. Mixed STID also leads pulse MAE (4.563). There is no robust reversal of model superiority.

B Equal-history diagnostic floor

For a pair with exactly equal histories, a deterministic history-only predictor must use the same output p for both arms. The triangle inequality gives

$$\frac{\mathrm{MAE}(p, y^B) + \mathrm{MAE}(p, y^F)}{2} \geq \frac{\mathrm{MAE}(y^B, y^F)}{2}. \tag{2}$$

The hindsight midpoint attains the bound but is not a usable predictor. This elementary constraint concerns equally weighted constructed pairs, not original-snapshot or real-traffic Bayes risk.

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Table 1: Center MAE with harmonized evaluation populations. STID entries are means over three training seeds; other configurations are deterministic. Event/control sample columns score all 12 steps, including samples without a future mark. The two variants are different constructed populations.

Variant	Configuration	All steps	Marked steps	Marked windows	Event samples	Controls
Generated	Last value	14.571	6.275	11.151	11.475	19.376
Generated	History mean	22.302	13.267	17.247	16.982	30.560
Generated	Linear drift	17.865	5.712	13.443	16.906	19.353
Generated	Ridge center	14.396	4.791	10.844	12.247	17.732
Generated	Ridge local-9	14.603	4.774	11.012	12.373	18.064
Generated	STID (no calendar)	11.499	4.823	9.013	9.673	14.334
Mixed	Last value	30.160	6.275	11.151	11.475	51.302
Mixed	History mean	40.532	13.267	17.247	16.982	67.179
Mixed	Linear drift	49.468	5.712	13.443	16.906	86.312
Mixed	Ridge center	30.813	5.021	11.921	12.344	51.711
Mixed	Ridge local-9	31.432	4.977	12.075	12.768	52.551
Mixed	STID (no calendar)	26.172	4.563	9.359	9.824	44.670

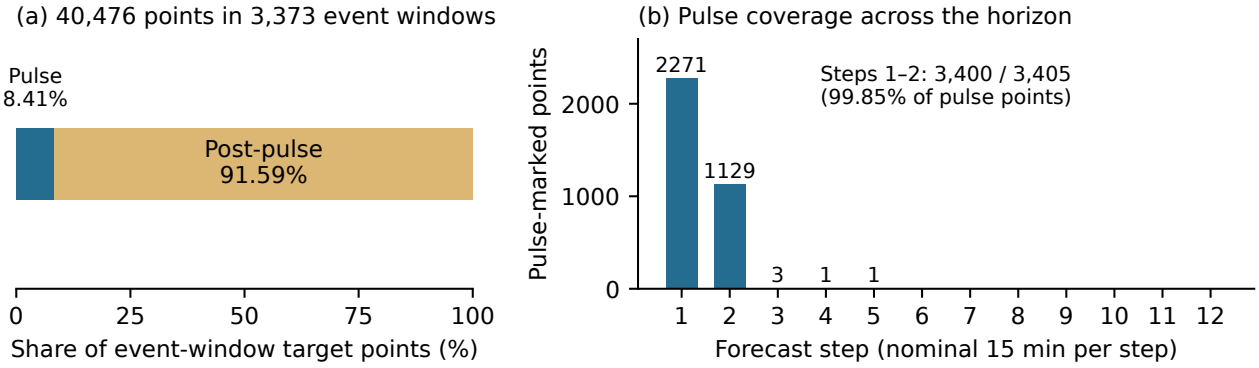


Figure 1: What the original event labels cover in both test partitions. Whole event windows are 91.59% post-pulse, whereas 99.85% of pulse-marked targets occur within two forecast steps. These are generator timing and scoring populations.

Table 2: MAE by generator phase in the original test partitions. Each variant has 3,405 pulse and 37,071 post-pulse points. STID uses three-seed means; phases have different horizon distributions.

	Generated		Mixed	
Configuration	Pulse	Post-pulse	Pulse	Post-pulse
Last value	6.275	11.953	6.275	11.953
History mean	13.267	17.323	13.267	17.323
Linear drift	5.712	17.934	5.712	17.934
Ridge center	4.791	12.932	5.021	13.016
Ridge local-9	4.774	13.071	4.977	13.483
STID (no calendar)	4.823	10.118	4.563	10.307

For 95 equal-history pairs, the balanced whole-window floor is 1.4599 versus errors 11.2908/18.0939/11.7453 for last value/history mean/linear drift. On 190 marked positions per arm, the floor is 2.3653. This constructed ambiguity neither proves identical-input conflicts in the original snapshots nor explains their event–control gap.

C Simple-predictor paired results

Full-operation MAE decreases in 104, 69 and 122 of 256 contexts for last value, history mean and linear drift (Table 3). History mean's median difference is positive (0.020) despite its negative mean. All three full-operation WMAPEs exceed background WMAPEs. The preserved four-arm results describe this construction and do not attribute the original event–control gap, whose controls and backgrounds differ.

All 95 future-starting pairs have equal histories and $A = 0$, yet history mean's full-arm MAE is 0.2428 lower entirely through target change. Timing-specific results and all crossed scores are retained with the artifact.

D Observed population sensitivity

The observational check fixes the first 512 node-order positions and January 2023 of TraffiDent version 8. Its five-minute flow channel makes 12-to-12 prediction a one-hour task. Complete finite, nonnegative windows stay within January 1–20/21–24/25–31 train/validation/test partitions; hourly cutoffs yield 201,210 training and 70,928 test windows. Three extrapolators and fixed ridge

Table 3: MAE on 256 fixed generated contexts, with identical context and background across arms. B: background; P: pulse only; R: recovery only; F: complete operation. Differences are computed before rounding. These contexts form a separate construction diagnostic, not additional independent traffic data.

Predictor	B	P	R	F	$F-B$	$F-P$
Last value	10.507	11.517	9.599	10.704	0.197	-0.813
History mean	17.019	17.215	16.225	16.820	-0.199	-0.395
Linear drift	10.546	15.591	10.489	15.316	4.770	-0.275

Table 4: Crossed-loss decomposition of the complete operation versus background, all 256 contexts and all future points. $D = T - A$ before rounding. Positive A means using the changed history reduces error on the full-operation target. These are components along the specified comparison path.

Predictor	Total D	Target T	Gain A
Last value	+0.1973	+0.1825	-0.0148
History mean	-0.1991	-0.1357	+0.0634
Linear drift	+4.7704	+0.8578	-3.9126

require no parameter search. Parsing, matching, exclusions and clock assumptions are in Appendix E.

Changing the label association. Nearest matching on freeway and direction within 0.5 miles, with smallest-ID ties, yields 193 future-event windows. Omitting direction gives 164, sharing 97 (Jaccard 0.373). Among 28,713 reports matched by both rules in full metadata, 14,789 change sensor. Fixed ridge event MAE is 30.321 versus 26.269 (Table 5); all four predictors retain their ordering and higher event MAE than for no-overlap windows. These are association rules, not physically verified affected sensors.

Changing forecast-horizon weights. Removing boundary, missing and overlapping reports leaves 91 event-aligned cases at 58 nodes: 422 target points intersect reported durations and 670 follow their ends. Both phases cover all twelve horizons, but active counts fall from 73 to 18 while post-end counts rise from 18 to 73. Equal-horizon weighting changes ridge's post-end-minus-active MAE from +0.875 to −0.466; last value and history mean retain smaller positive differences. This descriptive reference does not balance incidents or severity, and report resolution does not imply traffic recovery.

D.1 Eligibility and calendar controls

An all-road-type matching family permits last-value/ridge rank changes in January, but not February. This result depends on allowing different sensor types; all three STID seeds outperform the four original predictors throughout the allowed families. Appendix H specifies the rule; the artifact retains all ranges and witnesses. These choices are not equally credible substitutes for a forecasting task that specifies mainline sensors.

Eligibility before matching. TraffiDent's published forecasting task uses mainline sensors [11]. Filtering the complete metadata to that type before nearest matching retains 77 January and 273 February reports from the inspected cohorts, with a unique eligible nearest sensor for each. Thirty-eight old assignments violate this narrower contract. Ridge MAE is 32.992/33.134, versus STID ranges 31.724–31.854/31.599–31.853 in January/February. The apparent association ambiguity does not survive the stated mainline task; it is not evidence against that source task (Appendix J).

E External protocol and exclusions

TraffiDent bindings are Kaggle `gpxlcj/xtraffic` v8 and commit `b773335d31fa`. Verified ranges extract 512 nodes of January's $16972 \times 8928 \times 3$ tensor. Author code corroborates channels and a five-minute grid but uses another layout; full lineage, clock alignment and online availability remain unverified.

Of 38,072 January reports, 37,841 have usable matching metadata; all types remain. Positive durations define half-open intervals and other durations 24-hour exclusions. Matching searches all 16,972 sensors within 0.5 postmiles with minimum-ID tie resolution. Direction/freeway-only rules yield 717/792 reports, with 678/752 valid durations; 400 latter assignments differ in direction.

Complete finite, nonnegative windows give 201,210 training, 40,109 validation and 70,928 test examples. Ridge uses training-only scaling, an unpenalized intercept, regularization 1 and nonnegative clipping; metadata are evaluator-only. Test group memberships and exclusions are retained. Event cutoffs use $\lfloor t_{\text{report}}/5\text{min} \rfloor + 1$, including exact boundaries. Boundary, incomplete and overlap exclusions leave 91/178 primary and 80/167 freeway-only reports, with both phases covering twelve horizons.

F NET-V4 implementation and access

The local bundle retains the 49.6 MB base archive, errata and companions. `resource.py` offers nine executable replay entries and six forecast-scoring entries, including the separately identified MV task. Residual-target replay matches 330 aggregate and 5,632 context rows. MV arrays regenerate exactly from the retained profile library; legacy raw-source regeneration remains incomplete. The station checks require the exact upstream file, excluded from the bundle. Upstream notices apply separately. *Reviewer access will be added at submission.*

F.1 Fixed development partitions

The 3,373 event tests comprise 1,102 history-completed, 1,151 cutoff-spanning and 1,120 future-starting pulses. New predictions follow a fixed evaluation contract; earlier test exposure and shared profiles limit independence.

F.2 Baseline configurations

Last value repeats the final center value; history mean repeats the 12-step mean; linear drift extrapolates the last three-interval change.

Table 5: TraffiDent fixed-subset test results. Event-window MAE uses 193 direction-aware or 164 freeway-only windows; the direction-aware no-overlap group has 70,510 windows. The last two columns use a separate set of 91 isolated event-aligned reports and give post-reported-end minus reported-active MAE, before and after equally weighting the 12 common horizons. Different populations and physical units preclude direct comparison to NET-V4 scores.

	Regular-grid MAE			Event-aligned phase difference	
Predictor	Event, direction	No overlap, direction	Event, freeway only	Original weights	Equal horizon weights
Last value	33.911	19.204	28.950	+3.010	+0.745
History mean	43.139	23.948	35.676	+4.968	+0.523
Linear drift	69.488	37.791	63.187	+45.071	+33.080
Ridge center	30.321	16.989	26.269	+0.875	-0.466

Ridge center uses 12 center values; Ridge local-9 concatenates all nine histories. Both predict 12 center targets with training-only feature standardization and fixed regularization 1, without search.

STID has a 64-dimensional history embedding, 32-dimensional local-slot embedding and three MLP blocks. Calendar embeddings are disabled for lack of genuine hour/weekday inputs. Slots, including the repeated center, are local positions, not globally identified roads; no cross-node aggregation occurs. Training minimizes nine-slot L1, while center validation MAE selects checkpoints, so comparison with center-only ridge is not an architecture-only ablation.

STID uses seeds 1, 2, and 3, 30 epochs per seed, Adam with learning rate 10^{-3}, batch size 256, and gradient clipping at 3. Each variant has a training-only global mean and standard deviation. The model has 58,156 parameters. Runs used PyTorch 2.4.1 and one RTX 2080 Ti. Every model's final predictions are clipped at zero. The main NET-V4 evaluation uses four ridge and six neural fits. Four further ridge fits check input identity below; the post hoc selection check reuses only the original configurations (Appendix K).

F.3 Input identity sensitivity

Four fixed ridge fits compare center-pair slots 0/1 with deduplicated slots 0/2–8, preserving splits, scaling, regularization and clipping. Removing the copy changes overall MAE from 14.602861 to 14.582714 (generated) and 31.432328 to 31.427143 (mixed); center-pair scores are 14.416107/30.819166. Neural accuracy advantages remain. Correlation changes effective regularization, precluding an isolated spatial-information effect. STID is not retrained.

Generated-snapshot test centers all occur in training. Mixed-snapshot test data include 1,819 rows with unseen recorded center IDs, all observation-tagged controls; event test centers are all seen. This is not source-level or neighbor-level independence. Exact numerical matches to training occur for 102 mixed-test center histories and 74 full center sequences, but no full nine-slot sample. Such equality need not imply the same original source record. Profile construction pools source rows before nominal-year splitting; full source lineage remains unavailable.

F.4 Reproducibility checks

Target-free CPU replay of sixteen outputs differs by at most 4.89×10^{-4}. Generated ridge and STID seed 1 retraining reproduce weights and forecasts exactly with NaN test targets. Records are retained; cross-hardware equality is unproven.

F.5 Paired construction and crossed scoring

The 256 contexts are evenly spaced legacy event rows. Preserved functions, profiles, seed and random order exactly replay the 512-row two-arm prototype at float32 precision; pulse-only and severity-zero recovery-only arms add 512 diagnostic rows. Clipping is unchanged. All sixteen crossed losses and timing groups (85/76/95) are retained; these are development-exposed constructions.

F.6 Recovery-target replacement

For background v, event end e, final unclipped pulse value a and $q_t = \exp(-(t-e)/\tau)$, old recovery is $v_t + (a - v_t)q_t$. The single alternative is $v_t + (a - v_{e-1})q_t$. Here $\tau = \max(2, r/900)$ for response seconds r; clipping is unchanged. Before clipping their difference is $(v_t - v_{e-1})q_t$. The alternative preserves background variation and is the identity at zero pulse; neither law is established as physical truth.

Construction and scoring follow a fixed protocol, preserving original arms and random draws. Recovery starts at positions 12/13/14 in 85/76/95 contexts, leaving histories unchanged. Eleven saved configurations need no new inference. Independent scalar reconstruction verifies 330 aggregate and 5,632 per-context rows. These 256 new targets are developmental constructions; no parameter/context search or independent transfer validation is performed.

F.7 Source-bound observable-history probe

Seven January-frozen predictors score 2,492 February windows at 89 Mainline stations, daily 11:00–12:55, under background/pulse/zero-pulse-old-recovery/old-full/residual-full constructions. Fixed starts 12/11/6, duration 2, amplitude 0.12 and $\tau = 2$ precede scoring; clipping preserves zeros. For recovery-visible histories, old-law gain splits into zero-pulse $S = -1.3882$ and incremental-pulse $P = -12.5612$. Every seed/week has negative gains under both laws. Scalar construction and NumPy inference verify this exposed diagnostic; 747 NYC windows support structural checks only.

F.8 Fixed training-utility screen

Five ridge fits augment 201,210 January windows with scaling, pulse, zero-pulse old recovery, old full recovery or residual full recovery. Original/augmented weights are 0.5/0.5, regularization is 1 and starts 6/11/12 use seed 27003117; other parameters follow Appendix F.7. On 273 February reports, MAEs are 33.1343 unaugmented, 33.1372/33.3201/33.3875 for controls and 34.0414/33.7513 for full operations. Both full operations also worsen validation/noon error,

Table 6: Event minus control error. WMAPE differences are percentage points. Positive signs reverse the lower-event-error interpretation obtained from MAE.

	Generated		Mixed	
Configuration	MAE	WMAPE	MAE	WMAPE
Last value	-7.901	-1.851	-39.827	-10.008
History mean	-13.578	-4.118	-50.197	-11.004
Linear drift	-2.447	+4.001	-69.406	-19.415
Ridge center	-5.485	+0.181	-39.367	-9.252
Ridge local-9	-5.690	+0.075	-39.783	-9.163
STID (no calendar)	-4.661	-0.097	-34.846	-8.894

failing the fixed utility gate. Weighted replay verifies scores; this does not establish general augmentation failure.

F.9 Replay and paired interface

Eleven configurations pass 21,952 comparisons, including 392 decompositions and 6,272 crossed losses. Center inference agrees with full-model outputs. Missing neighbors exclude local-9 ridge. The interface separates 1,024 keyed histories from metadata. Context origins (162/46/48 train/validation/test) remain development-exposed; this is not an unseen evaluation.

G Normalization contrast details

Table 6 reports metric-specific event–control differences.

H Association support and endpoint checks

Reports with identical complete candidate sets share one sensor choice. For per-report loss differences $d(i, j)$, extrema sum each group's minimum/maximum of $\sum_{i \in I_k} d(i, j)$ and divide by report count. This is elementary separability, not a confidence interval or new optimization theorem. Related uncertain-data work includes certain predictions [14] and possible-model analysis [15]. The last-value/ridge interval is $[-0.536, 2.933]$ in January and $[0.991, 3.405]$ in February. The complete prior table remains in the supporting artifact.

The 178 January reports are selected by minimum-ID matching before completing candidate sets. Of 145 support nodes, 63 are newly acquired; none enters training. Boundaries, incomplete candidates and overlaps exclude 2/67/29 reports, leaving 80 with 123 edges. The 48 disjoint sets permit 5,971,968 shared choices and globally consistent endpoint priorities. Unknown-duration overlaps use 24-hour exclusions. Independent source reconstruction and endpoint checks agree within 1.5×10^{-14}, without establishing association truth.

I Temporal extension and metadata sensitivity

February's node-major float32 tensor is $16972 \times 8064 \times 3$; 260 blocks contain 25,159,680 bytes. Matching precedes acquisition; predictions precede scoring. Overlap checks include prior-month carryover and 24-hour unknown-duration exclusions. There are 166 edges on untrained nodes. The 95 disjoint sets permit global-priority endpoints.

External STID uses history/center embeddings of dimensions 64/32, three residual MLP blocks, dropout 0.15 and 57,900 parameters. Training uses normalized L1, Adam 10^{-3}, batch 512, gradient clipping 3 and 30 epochs for seeds 1/2/3. January training histories determine global scaling. Nonnegative-clipped center validation selects epochs 27/29/30 with MAE 15.175/15.187/15.158. January test and February observations enter neither fitting nor selection; no grid-wide or state-of-the-art comparison is claimed.

February replay differs by at most 4.0×10^{-4}; two-month endpoints by 3.6×10^{-14}. Ambiguous groups mix road types. These checks establish neither sampling uncertainty nor association truth.

J Mainline target-scope check

TraffiDent's main text names 561 San Bernardino mainline sensors; Table 3 and Appendix A.8 instead name Monterey. This source inconsistency remains unresolved. Our mainline eligibility rule is explicit, not a reconstruction of that historical experiment. This post hoc check retains 355 reports and seven frozen predictors. Mainline eligibility precedes matching without fallback; 10 January and 28 February old assignments violate it. Metadata fixes 13 additional blocks before scoring. After missing-sensor and overlap exclusions, 77/273 reports remain. Inference verifies scores within 4.0×10^{-4}; location tokens do not identify the truly affected sensor.

K Fixed validation-selection use check

Four validation criteria compare six configurations without fitting; STID averages seed losses and the balanced criterion weights pulse/recovery equally. Generated pulse selection worsens both event and post-pulse MAE, as recorded with full scores in the artifact. Center/local ridge wins 341/21 of 362 segment omissions; other criteria always choose STID. All mixed criteria choose STID in 182 omissions. Selection precedes scoring, but prior exposure precludes independent confirmation.

L Observed calendar-pair protocol

Let $c = \lfloor t_{\text{report}}/5\text{min} \rfloor + 1$; bin $c - 1$ contains the report. The pre forecast uses history $[c-25, c-13)$ and target $[c-13, c-1)$, strictly before that bin. The post forecast uses $[c - 12, c)$ and $[c, c + 12)$. Each episode therefore requires 37 finite, nonnegative flow values in $[c - 25, c + 12)$. These two forecasts share source points and are not independent repeats.

Controls shift cutoffs by $\pm 7, \pm 14, \pm 21$ days within February, retaining all complete, nonoverlapping episodes. Unknown or nonpositive durations use 24-hour exclusions. Four overlapping cases and one without controls leave 268: 58/69/61/80 by seven-day block. Forty-one have two controls and 227 have three, equally weighted within equally weighted cases. Pairing uses no future traffic or best-control selection.

The 1,031 episodes contain 2,062 origins and 37,548 unique source points; 599 repeat. Seven predictor replays match 792 aggregate, weekly and station-omission scores within 5.0×10^{-11}. Fixed histories, targets, pairs and forecasts remain available. Replay checks saved cohorts, not selection reconstruction or independent incidents.

The positive-G gate covers the aggregate and all four weeks per seed. A separate positive-pre/post-gap but nonpositive-change gate fails. Neither is a significance test. Development exposure, nonrandom selection, assumed clocks and unresolved online availability, sensor identity and processing lineage limit interpretation.

M Fixed forecast-risk utility screen

Seven frozen predictors calibrate on 74 cases and 7,107 hourly controls from January 25–31; 86,514 of 89,846 source points recur. February retains the exposed 268/763 comparison. For $s = \max_h |y_h - p_h|$ and $a = 1 + \mathrm{MAE}_{\mathrm{pre}}$, U calibrates s, S calibrates s/a, and R/RS additionally stratify by report status. Level-0.90 inverse-CDF quantiles use no interpolation/correction; scaled radii multiply by a. This standard grouping [20] supplies no coverage guarantee.

Bands are $[\max(0, p_h - q), p_h + q]$; pinball loss is $0.9(s - q)$ for $s \geq q$, else $0.1(q - s)$. Case/control groups have equal weight. RS must beat U/S by 1% per seed and each week, preserve coverage and limit control cost to 0.5%. Gains over S are 2.869/0.299/0.763%; every cost gate fails. Mixed widths are 291.782/235.026 for RS/U. All predictions and 28 calibration replays are verified; no subsequent search occurs.

N Public-release audit and lookup replay

The source digest binds the 865,971,481-byte v1 file; code matches the pinned Git tree. A symbolic parser avoids pickle execution, while restricted native decoding verifies 22,340 arrays and metadata. Duration serialization was repaired without changing calendar results; neither that field nor published normalization statistics proves leakage.

Tracing uses CSV row order locally; the `order` column holds global indices. All 13 source stations have 8,928 finite integer flows; released inverse histories deviate by at most 1.90×10^{-5}. Independent matching reads original source bytes without permutation search.

Training-only lookup replays all 19,951,104 predictions exactly and metrics within 1.43×10^{-14}. The 184 repeated training keys have no target conflicts. Validation/test RMSE changes from 30.796/30.650 to 28.535/28.739. Dates and original neural-model inflation remain unresolved.

O Fixed-query target-support intervention

TargetSupport/WholeTraffic remove 1,244/97 training rows, preserving test queries and weights. Byte checks cover 11,170 history/target arrays, remaining matches and removal witnesses. Fixed test groups contain 93 full-traffic-reuse, 466 other-target-reuse and 280 no-original-target-reuse queries. All six conditions' MAE/RMSE and query losses remain available, including unfavorable RMSE changes.

For Original versus TargetSupport, aggregate MAE changes by −0.000607 (−0.0042%), while RMSE rises from 26.9344 to 27.0192. The 93 full-traffic-reuse queries worsen from 15.6667 to 15.8545 MAE; the 280 no-original-target-reuse queries improve from 14.0960 to 14.0248. These offsetting finite changes are conditional on the changed training composition; they do not identify a causal memory effect.

Augmented QR independently verifies 29,962,368 predictions within 1.17×10^{-10} and 48 group metrics. Six research/six reproduction configurations each contain 496 stationwise systems. Predictions precede scoring; masks use exposed targets. Hour-distribution total variation is 0.0527, so equal training size does not isolate composition.

The subsequent projection audit compares same-position 12-step histories and futures, with a stricter variant also requiring identical 24-step calendar bytes. Of 416,144 query/node cells, 341,680 have varying futures. TargetSupport retains 76,542 calendar-aware paired matches: 30,685 varying, 45,662 constant near-zero and 195 other constant targets. Byte equality defines every match; tolerance 0.001 applies only to the inverse-flow zero category. Independent sorted-record membership and byte comparisons verify all positive witness groups, without fitting a predictor.

On the remaining varying paired matches, Original and TargetSupport MAEs are 8.3067 and 8.4099; elsewhere they are 15.0458 and 15.0369. These target-defined post hoc groups have different difficulty and do not identify a contamination effect. Complete masks, memberships and all six conditions' counts are retained. Two damaged Chinese protocol texts have dated retrospective reconstructions; sealed code and parameters remain verifiable.

Local lookup verification. Adding historical calendars yields MAEs 13.3843/15.3356 on Original/TargetSupport. The latter has 53 conflicting-future hits and uniquely recovers 30,674 varying futures. Independent source decoding and sorted-record search verify all 19,974,912 predictions exactly and scores within 1.3×10^{-14}. All four conditions, ambiguous hits and constant-target groups remain in the companion.

Input-exclusion feasibility. Four fixed view/key conditions retain all queries and nodes, without fitting or parsing query targets; TargetSupport retains its prior target exposure. Flow-only exclusion leaves 0–3,807 Original and 0–2,660 TargetSupport rows per station (medians 3,709/2,591.5), with the same 51 empty positions. With historical calendars, ranges are 2,369–3,812 and 1,679–2,665; residual flow-only hits number 75,284/75,167 across all 496 positions. Declared-key hits vanish by construction, not as a new algorithmic result. A post-result check finds that common-row intersections exactly select calendar keys absent from queries, not recovered dates. Independent checks verify all 7,755,456 mask entries, earliest removal witnesses and memberships. Nonempty views do not establish adequate training support; empty views preclude fitting a station independently from that view alone.

P External target-construction probe

CiK commit `73f46016` postdates publication; TrafficFresh `72f9682a` supplies a 1 MiB prefix (7,736 complete records). Both stations succeed on 32 seeds with identical method/predictor code, nine response cells and eight distinct rules. Scalar checks verify 1,536 targets exactly, 6,912 forecasts within 1.78×10^{-15} and 1,206 scores within 4.45×10^{-16}. Mean-profile reversal holds under all three weightings at both stations. Last-day reversal holds only for full-horizon/pooled-event weighting at the first station, and all three at the second. Predictions precede scoring; the substituted loader precludes claiming published CiK reproduction.

Q Original-split reuse and bounded-target addendum

These two studies were fixed on 9 October 2026 after the original structural and endpoint results had been seen. Their dated protocols are execution-before-scoring records, not independent preregistrations. The original arrays, split indices, eleven paired forecast configurations and two clipped endpoint targets are unchanged.

Q.1 Core lookup consequences

The primary key is the byte representation of the twelve float32 center-history values; the scope comparison keys on all 12×9 historical values. For a test history x, let $J(x)$ contain every matching training row under the declared key. A hit predicts the float64 mean of the center futures of all rows in $J(x)$; a miss retains its fixed fallback. LastValue is primary. Each of the three saved STID test predictions provides a secondary miss fallback. No test target, event tag or source identifier selects a training future. The two snapshots, two keys and four fallbacks yield sixteen sealed rule output files, with zero new fits. An independent tuple-key path verifies the matched sets and predictions. Prediction sealing precedes scoring, but does not undo earlier development exposure.

The mixed center key has 102 test hits; 74 have some exactly repeated training center future, and 67 have conflicting training futures. These categories are not a partition. All hits are controls. LastValue MAE changes from 30.1600868 to 30.1239257 over all 6,354 queries (0.119897%). On the 62 varying-target hits it changes from 8.7892605 to 2.7272780; on the 40 constant-target hits it worsens from zero to 3.6518759. On conflicting-future hits it worsens from 2.0060324 to 3.6727056. These groups use targets only for retrospective scoring, not prediction. Event loss and every missed-query prediction remain unchanged. With the three STID miss fallbacks, overall MAE gains are 0.114090%, 0.157669% and 0.125066%; the mean of their losses changes from 26.1723193 to 26.1376813. The generated snapshot has no hits under either key; neither snapshot has full-nine-slot hits. Full witnesses, scores and seals are retained in `saved_lookup`. These are fixed-rule consequences, not estimates of what the original neural model memorized.

Q.2 Complete bounds on one target segment

For frozen clipped targets y^0, y^1, let $y^\alpha = y^0 + \alpha d$, where $d = y^1 - y^0$ and $\alpha \in [0, 1]$. Histories and forecasts remain fixed. This segment interpolates target arrays after clipping; it does not vary a generator parameter or define a physical uncertainty range. For each scalar prediction p_k, $|p_k - y^0_k - \alpha d_k|$ changes slope only at $(p_k - y^0_k)/d_k$ when $d_k \neq 0$. Collect all interior crossings for both background- and full-history forecasts, plus 0 and 1. Each loss and their differences A, D, T are linear between these sorted knots, so minima and maxima occur at knots. Linear interpolation locates isolated sign changes. This is standard absolute-loss geometry; a numerical zero tolerance of 10^{-9} is declared.

The runner accumulates slope jumps $2|d_k|/m$ for m scored scalar elements. Direct NumPy loss evaluation independently checks endpoints, extrema, a fixed selection of knots and four non-knot positions. A known finite example with negative response at both endpoints and positive response internally verifies that the routine can detect such a reversal. Saved results retain all eleven configurations, both three-seed loss means, and three declared cohorts: all 256 contexts, 48 test-origin contexts and 95 future-starting contexts. Seed aggregation averages losses rather than forecasts. All existing deterministic configurations have zero response on the 95 equal-history pairs.

The predeclared full-cohort hidden-reversal hypothesis is unsupported. For mixed STID's loss mean, response is adverse throughout the segment, while the diagonal error difference crosses zero at $\alpha = 0.816080569$. Mixed seed 3 has slightly positive response at the replacement endpoint; it is retained rather than removed from the aggregate. The complete curves, knot locations, extrema and source/forecast seals are in `saved_target_segment`. A reader's same-format forecast file can be scored without fitting. The scorer computes conditional losses for the supplied file and does not certify that a submitted forecast was generated by a deterministic history-only model.

Q.3 Offline replay boundary

The `netv4_checks` replay reads fourteen hash-bound files from the unchanged original archive, recomputes the sixteen lookup rules, and scores the saved paired predictions. It requires neither the parent development tree, network access nor a GPU. It checks both regenerated summary dictionaries against the saved results. This verifies reproducibility of the supplied objects, not raw-source regeneration or new independent scientific evidence.

R Resource increment and intended research decisions

The focused comparison uses four previously acquired works and pinned code versions. General test tools can express these comparisons; their implementation is not a new testing method. Table 8 instead identifies the packaged input/target objects, saved outcomes and research decisions. The source-bound artifact map names the relevant files and commands. This comparison establishes neither exhaustive priority nor missing functionality in the related tools.

Prediction researchers. Submit fixed center or joint-input forecasts, retain their accuracy, and inspect response on common targets. A lower diagonal loss does not establish a helpful response. Positive response also does not replace accuracy: the supplied history-mean example is helpful but inaccurate.

Joint-input model users. The MV views keep the center fixed while changing other histories and vice versa. They reveal conditioning and timing dependence for the supplied models; the cases do not establish measured road effects or an attribution law. Zero responses from center-only configurations and equal-history future cases remain structural controls.

Resource curators. State the prediction input key before treating record uniqueness or an exclusion count as a support claim. The supplied witnesses connect matches to lookup consequences and remaining training rows. Source identity, chronological independence and statistical adequacy require evidence beyond numerical equality. No experiment demonstrates community adoption or saved researcher time.

Table 7: Complete loss ranges over the declared target segment, all 256 contexts. Three-seed summaries average losses. These finite ranges are not confidence intervals. No full-cohort configuration with same-sign endpoint response has an internal opposite-sign response.

Configuration	Response A range	Error difference D range	Response status
Last value	[-0.040664, -0.014768]	[0.197273, 0.978434]	uniformly adverse
History mean	[0.059556, 0.063384]	[-0.199090, 0.555694]	uniformly helpful
Linear drift	[-3.912564, -3.819872]	[4.714406, 4.770387]	uniformly adverse
Generated / Ridge	[-0.891191, -0.723684]	[0.987727, 1.462520]	uniformly adverse
Generated / STID seed 1	[-0.213598, -0.149256]	[0.031911, 0.642374]	uniformly adverse
Generated / STID seed 2	[-0.242871, -0.158324]	[0.168731, 0.742510]	uniformly adverse
Generated / STID seed 3	[-0.231005, -0.154967]	[0.131434, 0.690417]	uniformly adverse
Mixed / Ridge	[0.089693, 0.163495]	[-0.045436, 0.580631]	uniformly helpful
Mixed / STID seed 1	[-0.154979, -0.084139]	[-0.577277, 0.120930]	uniformly adverse
Mixed / STID seed 2	[-0.126504, -0.090586]	[-0.603597, 0.098674]	uniformly adverse
Mixed / STID seed 3	[-0.042871, 0.005510]	[-0.507613, 0.196443]	sign or zero sensitive
Generated / STID loss mean	[-0.229158, -0.154182]	[0.110692, 0.691767]	uniformly adverse
Mixed / STID loss mean	[-0.108118, -0.056405]	[-0.562829, 0.138683]	uniformly adverse

Table 8: Established capabilities and the specific NET-V4 objects available for reuse. Every contrast retains its own task and source scope.

Related resource	Existing capability	Concrete NET-V4 object and reader decision
CheckList [16]	Functionality, invariance and directional tests, reusable test suites.	Keyed $B/P/R/F$ center histories and crossed targets; submit a fixed predictor and distinguish lower event-arm error from helpful same-target response.
Context is Key [17]	Contextual forecasting, text ablations, region scoring and fixed configurations accepting given targets.	Complete $B/P/R/F/C/N$ histories, four timings and two declared targets; reuse one forecast file for conditional input and timing decisions. Two hourly CiK probes remain separate supplements.
TraffiDent [11, 19]	Observed traffic/incident data, post-incident forecasting comparisons and same-station prior-day controls.	Fixed case/control memberships, strictly pre-report targets and saved January-frozen forecasts; inspect a descriptive observed comparison without treating it as validation of synthetic incident laws.
IGSTGNN release [21]	Traffic/calendar/incident records and a documented split script.	Source-bound history/future witnesses, hit/miss losses and per-station exclusion masks; inspect input-unit reuse and retained support. The external file is required for source-audit replay.

S Complete-input MV task and all frozen configurations

The separate MV protocol was written before generation, prediction and scoring. It uses 102 retained weekly profile arrays; original profile identifiers group the conditional results but do not restore their raw-source pool. Neighbor indices follow fixed numeric profile similarity. Reference week, hourly coordinates, severity cycle, attenuation parameters and timings are prescribed. The full resource includes source profile arrays, context-to-profile provenance, pinned function bodies, code and configuration. Original snapshots, weights and the center task are unchanged.

There are 2,856 backgrounds and four scenarios per background. History-completed, spanning, early and late groups have $(s, d) = (8, 4), (10, 8), (14, 4), (18, 6)$. Every scenario has $B/P/R/F$ plus functional histories C, N on the same full target. Center duplication is exact. No model receives IDs, parameters, phase masks or future targets through the adapter. Pulse support at forecast steps is 2,856, 2,856, 5,712, 5,712, 5,712, 5,712, then 2,856 at each of steps 7–12. This increases declared diagnostic coverage; it is not a correction to the original sampled population.

Thirteen frozen model configurations produce 68,544 center forecasts each. There are zero new fits, and all configurations are retained. All center-only models exactly preserve predictions under other-sequence changes. Both future-starting groups have identical complete histories in all six views and zero response for all models. Independent direct losses verify 5,408 reported numbers to within 1.78×10^{-15}. Reconstruction compares 16,702,704 array elements exactly. A coupled reader adapter runs with input arrays and weights only, and reproduces saved forecasts exactly before evaluation.

Generated and mixed joint Ridge have harmful profile-group mean other-response on 57 and 60 of 102 profiles, respectively, under both target laws; 45 and 42 profiles improve. These are grouped finite outcomes. They provide no station-population significance, realistic propagation guarantee or inference that the old fitted models used repeated traffic. The score's response gate compares conditional input views under the prescribed targets. All phases, laws and profile-group losses are retained in machine-readable CSVs.

T Neural joint-input resource-use validation

One fixed architecture addresses the absence of a neural predictor with actual cross-sequence aggregation. Graph WaveNet [7] uses official commit `6b162e80c59a1` (full binding in the artifact). Three two-dimensional-kernel Conv1d declarations are replaced mechanically by Conv2d for PyTorch compatibility. Adaptive adjacency, second-order graph convolution, four two-layer blocks and widths 32/32/256/512 remain; input/output channels are 1/12 and dropout is 0.3. Learned nine-slot connections are not measured road adjacency.

Each original snapshot supplies three primary fits, seeds 1–3. The fixed recipe uses training-history global scaling, nine-slot L1, Adam at 0.001, weight decay 0.0001, batch 256, clipping norm 5 and 30 epochs. Validation-center MAE selects each checkpoint. Original

Table 9: MV response for all thirteen frozen configurations, legacy target. Columns are finite conditional losses, not independent incidents or confidence bounds.

Configuration	Pooled A	Completed A	Spanning A	Early A	Late A	Spanning $A_{N\|C}$
LastValue	+0.5775	-1.5064	+3.8163	+0.0000	+0.0000	+0.0000
HistoricalMean	+0.1980	+0.4194	+0.3725	+0.0000	+0.0000	+0.0000
LinearDrift	-7.2938	-25.4307	-3.7446	+0.0000	+0.0000	+0.0000
Gen / Ridge center	-1.5339	-8.2874	+2.1517	+0.0000	+0.0000	+0.0000
Gen / Ridge joint	-3.1866	-14.2114	+1.4650	+0.0000	+0.0000	-0.0886
Gen / STID s1	-1.4837	-8.0504	+2.1156	+0.0000	+0.0000	+0.0000
Gen / STID s2	-1.7685	-9.6135	+2.5394	+0.0000	+0.0000	+0.0000
Gen / STID s3	-2.8469	-14.1135	+2.7257	+0.0000	+0.0000	+0.0000
Mix / Ridge center	-1.2326	-8.4615	+3.5310	+0.0000	+0.0000	+0.0000
Mix / Ridge joint	-1.3604	-8.7507	+3.3090	+0.0000	+0.0000	-0.0823
Mix / STID s1	-2.2036	-11.2700	+2.4558	+0.0000	+0.0000	+0.0000
Mix / STID s2	-2.6892	-13.0120	+2.2552	+0.0000	+0.0000	+0.0000
Mix / STID s3	-2.7014	-12.9774	+2.1719	+0.0000	+0.0000	+0.0000

Table 10: MV response for all thirteen frozen configurations, residual target. Columns are finite conditional losses, not independent incidents or confidence bounds.

Configuration	Pooled A	Completed A	Spanning A	Early A	Late A	Spanning $A_{N\|C}$
LastValue	+0.6215	-1.3391	+3.8251	+0.0000	+0.0000	+0.0000
HistoricalMean	+0.2048	+0.4488	+0.3705	+0.0000	+0.0000	+0.0000
LinearDrift	-7.0683	-24.8278	-3.4456	+0.0000	+0.0000	+0.0000
Gen / Ridge center	-1.2820	-7.4857	+2.3578	+0.0000	+0.0000	+0.0000
Gen / Ridge joint	-2.9168	-13.3695	+1.7021	+0.0000	+0.0000	-0.1149
Gen / STID s1	-1.2610	-7.2801	+2.2360	+0.0000	+0.0000	+0.0000
Gen / STID s2	-1.5488	-8.8589	+2.6638	+0.0000	+0.0000	+0.0000
Gen / STID s3	-2.6456	-13.3031	+2.7207	+0.0000	+0.0000	+0.0000
Mix / Ridge center	-1.0186	-7.6811	+3.6068	+0.0000	+0.0000	+0.0000
Mix / Ridge joint	-1.1351	-7.9410	+3.4005	+0.0000	+0.0000	-0.0799
Mix / STID s1	-1.9794	-10.4247	+2.5069	+0.0000	+0.0000	+0.0000
Mix / STID s2	-2.4612	-12.1784	+2.3335	+0.0000	+0.0000	+0.0000
Mix / STID s3	-2.4784	-12.1451	+2.2313	+0.0000	+0.0000	+0.0000

test targets are excluded from fitting; MV targets are not used for training, tuning or selection. All six forecasts are sealed together before decoding MV targets. This retains prior development exposure; it is not independent blind evaluation.

All six primary configurations have helpful total and harmful conditional other-response on spanning cases under both laws (Figure 2). The four three-seed means give error increases 2.55–3.54%, with helpful profile groups retained. All six also have adverse pooled but helpful spanning response. These fixed constructions establish a resource-use example, not a general claim that joint traffic models fail.

Actual C/F and B/N forecasts differ, with maximum spanning C/F changes 15.11–175.86. Gradients from 32 evenly spaced spanning contexts per model are nonzero for other inputs. Equal-history future-starting forecasts differ by at most 2.45×10^{-4}, below the fixed 10^{-3} tolerance. A separate loss calculation checks 240 primary pooled/spanning values within 5.33×10^{-15}. The history-only CPU adapter checks 822,528 values within 4.89×10^{-4} of CUDA forecasts.

The initial FP32-layout run completed Generated seed 1 while a process-permission delay left seed 2 partially trained. Both are retained. All six primary configurations then use FP32 channels-last storage, a computational layout fix without architecture or recipe changes. The completed superseded run is separately scored and excluded from the three-seed means: its spanning other-responses are −1.3303/−1.4574 for legacy/residual. There are seven completed fits and one interrupted attempt, not eight independent replications. Code versions and raw logs are preserved.

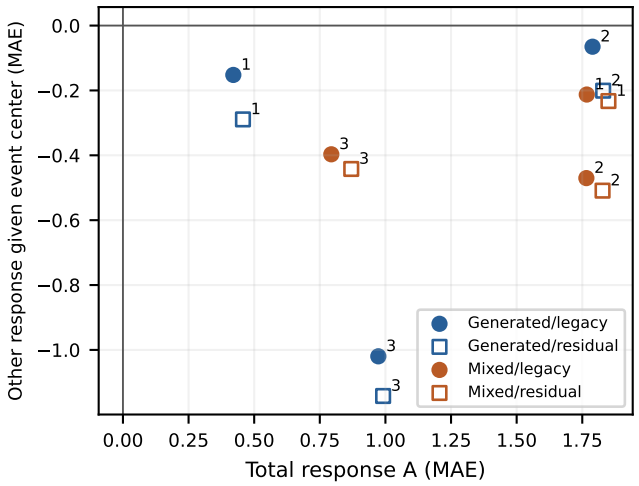


Figure 2: All primary neural seeds and both target laws on spanning cases. Responses are finite conditional losses, without population confidence intervals.

Original-test GWN center MAE is 11.1212 ± 0.1712 Generated and 27.8725 ± 0.4359 Mixed (seed standard deviations), compared with STID means 11.499 and 26.172, respectively. MV spanning legacy MAE is 15.1102 and 14.4594, versus STID seed means 13.1462 and 13.1855. GWN therefore does not provide a universal accuracy improvement. All original-test predictions, timing/phase losses, profile groups, structural checks and technical sensitivity remain available.

U History-only proximity and construction scope

This explanatory check asks whether the spanning response disagreement persists on histories satisfying one declared proximity rule. Earlier outcomes were development-exposed. The policy was fixed for this analysis, not independently preregistered. No model was fitted, no forecasts rerun, and no new scenarios or observations created. The resource adds one proximity analysis and one saved-forecast subset scoring run.

Each original training reference remains separate. Flatten the ordered 12×9 history, including the center copy, and scale all values by that reference's training-global mean and population standard deviation. Distance is standardized RMS to the nearest complete training history, found by exhaustive float64 search. The threshold is the original validation-to-training distance's 95th percentile with linear interpolation: 0.246640 for Generated and 0.536314 for Mixed. Include a scenario only when its B, C, F histories all qualify. Both response comparisons consequently satisfy the same rule. Selection is saved before subset outcomes are read. No metric, threshold, model or subgroup search follows.

All 11,424 scenarios receive distances; the primary population is the 2,856 spanning cases. The fixed descriptive coverage gate requires at least half of these cases and at least 80% of 102 profiles each retaining 14/28 cases. This is a coverage policy, without a significance or independent-incident interpretation. Mixed retains 2,631 cases (92.12%), represents all 102 profiles and qualifies 96 profiles (94.12%). Generated retains 2,095 cases (73.35%), represents 93 profiles and qualifies only 74 (72.55%); its profile gate fails.

Mixed's legacy losses are $\mathcal{L}_{CF} = 11.3953$, $\mathcal{L}_{FF} = 11.7314$; residual losses are 12.1053 and 12.4715. Conditional error therefore increases by 2.95% and 3.03%. Each law has 10 helpful and 92 harmful profiles under seed-mean losses. Generated seed 1 loses helpful total response under both laws; seed 2 has helpful conditional other-response under the legacy law. Its small positive row-mean A also becomes negative when represented profiles receive equal weight (Table 13). Thus the extension is supported only for Mixed under this rule; full-cohort findings are not replaced by this subset.

The nearest-reference source composition is descriptive. Among the 2,631 included Mixed cases, nearest observation-tagged control counts are 1,105 for B, 1,100 for C and 1,099 for F. Generated event counts are 1,466 for B and 1,473 each for C and F. Bootstrap control counts are 60, 58 and 59 in the same arm order. Source tags describe the frozen search's returned neighbour, without identifying a unique physical record. Proximity to this mixed constructed reference cannot validate independently observed incidents, road adjacency, the target law or parameter calibration. The rule establishes neither equal distributions nor density support throughout the input space. All full/included/excluded seed and profile losses, timing coverage and source counts accompany the artifact.

Sixteen evenly spaced queries per reference match independent CPU exhaustive distances exactly. All twelve full seed/law loss rows reproduce their frozen values within 3.56×10^{-15}. One portable re-execution extracts only the original history/split archive members and invokes the unchanged audit; it reproduces every selection array and 14,544 CSV entries. This is a mechanical reproduction of the same analysis, not independent scientific confirmation. Construction rationales and the distinction between retained quantities and prescribed parameters are documented in the audit component.

V Supplementary CiK target details

V.1 A fixed-input probe of a separate traffic generator

The pinned Context is Key traffic-decrease generator [17] selects an event interval $I = [s, s + d)$ and a multiplier $r \in \{0.1, \ldots, 0.5\}$. Within that interval it sets the target to rv_s, where v is the unmodified future, and describes a percentage of usual traffic. We test one alternative, rv_t, which retains temporal variation. Both are hypothetical target laws; neither is asserted to be physical truth or a correction of the author's intended task.

The pre-scoring source rule selects the earliest eligible run at sensor 423339 (1,834 hours). After seeing its results, we freeze a sequential check on the sole remaining prefix station, 10130110, using its earliest eligible run (1,653 hours). Both use the original generator method and window helper, seeds 0–31, and a disclosed fixed-series loader. The 168-hour history and 24-hour future differ from NET-V4's 12-to-12 task. Three base forecasts—last value, last day and a seven-day mean daily profile—use no response, anchored response rq_s, or pointwise response rq_t within I. Histories, scenarios and predictions remain fixed across targets; no rules are retuned.

The mean-profile comparison reverses at both stations (Table 15): anchored minus pointwise error changes from −0.04602 to +0.07727, and from −0.02627 to +0.00651, respectively. Last-day ordering reverses only at the second station under the primary measure; last-value responses coincide at both. Targets change at 108/144 and 101/138 event positions; outside-event targets stay equal. These are dependent constructed cases, without a significance or station-population claim. All no-response, full-horizon and pooled-event scores remain in the companion. The result supports bounded diagnostic reuse, not a new algorithm, physical target truth or CiK's published-model ranking (the technical companion).

Table 11: All primary Graph WaveNet checkpoints, legacy target; the superseded layout run is kept separately. No independent-incident inference.

Configuration	Pooled A	Completed A	Spanning A	Early A	Late A	Spanning $A_{N\|C}$
Gen / GWN s1	-2.2318	-9.3476	+0.4205	+0.0000	+0.0000	-0.1522
Gen / GWN s2	-3.0650	-14.0490	+1.7891	+0.0000	+0.0000	-0.0651
Gen / GWN s3	-3.0748	-13.2723	+0.9729	-0.0000	-0.0000	-1.0198
Mix / GWN s1	-2.6538	-12.3825	+1.7671	-0.0000	+0.0000	-0.2126
Mix / GWN s2	-1.9030	-9.3776	+1.7656	-0.0000	-0.0000	-0.4703
Mix / GWN s3	-2.8434	-12.1675	+0.7938	+0.0000	-0.0000	-0.3969
Gen / layout sensitivity s1	-3.7764	-16.9405	+1.8349	+0.0000	+0.0000	-1.3303

Table 12: All primary Graph WaveNet checkpoints, residual target; the superseded layout run is kept separately. No independent-incident inference.

Configuration	Pooled A	Completed A	Spanning A	Early A	Late A	Spanning $A_{N\|C}$
Gen / GWN s1	-2.0755	-8.7593	+0.4572	+0.0000	+0.0000	-0.2893
Gen / GWN s2	-2.9278	-13.5406	+1.8296	+0.0000	+0.0000	-0.2007
Gen / GWN s3	-2.9236	-12.6855	+0.9912	-0.0000	-0.0000	-1.1421
Mix / GWN s1	-2.4214	-11.5354	+1.8497	-0.0000	+0.0000	-0.2327
Mix / GWN s2	-1.7286	-8.7416	+1.8273	-0.0000	-0.0000	-0.5089
Mix / GWN s3	-2.6312	-11.3944	+0.8698	+0.0000	-0.0000	-0.4421
Gen / layout sensitivity s1	-3.6050	-16.2592	+1.8394	+0.0000	+0.0000	-1.4574

Table 13: Fixed-proximity spanning subset, three-seed loss means. Row weights reflect retained case counts; equal-profile weights give each represented profile one vote. Generated fails the coverage gate and its total-response sign changes with weighting.

Reference / target	Cases	Row A	Row $A_{N\|C}$	Profile A	Profile $A_{N\|C}$
Generated / legacy	2095	+0.0614	-0.1959	-1.3121	-0.3087
Generated / residual	2095	+0.0345	-0.2634	-1.3660	-0.3671
Mixed / legacy	2631	+0.7231	-0.3361	+0.4350	-0.3070
Mixed / residual	2631	+0.7821	-0.3662	+0.4929	-0.3349

Table 14: Every primary checkpoint on the fixed-proximity spanning subset. Positive total and negative conditional other-response coincide in all six Mixed rows and only three Generated rows.

Reference / seed / target	A	$A_{N\|C}$
Gen / s1 / legacy	-0.4820	-0.0468
Gen / s1 / residual	-0.4976	-0.1149
Gen / s2 / legacy	+0.5554	+0.0285
Gen / s2 / residual	+0.5317	-0.0404
Gen / s3 / legacy	+0.1109	-0.5694
Gen / s3 / residual	+0.0695	-0.6347
Mix / s1 / legacy	+0.9302	-0.2164
Mix / s1 / residual	+0.9972	-0.2325
Mix / s2 / legacy	+0.9722	-0.4359
Mix / s2 / residual	+1.0237	-0.4717
Mix / s3 / legacy	+0.2670	-0.3559
Mix / s3 / residual	+0.3255	-0.3945

Table 15: Case-equal event MAE, 32 constructions per station. Columns cross two hypothetical target laws with anchored/pointwise forecast responses. The second station is a sequential robustness check.

	Plateau target		Pointwise target	
History rule	Anchor	Pointwise	Anchor	Pointwise
Sensor 423339 (initial probe)				
Last value	.8263	.8263	.5429	.5429
Last day	.5322	.4894	.5635	.3565
Mean daily profile	.4696	.5156	.3458	.2685
Sensor 10130110 (fixed replication)				
Last value	.2116	.2116	.1753	.1753
Last day	.0969	.1320	.0859	.0701
Mean daily profile	.1093	.1356	.0826	.0761

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